



Leukocytes

(White Blood Cells, WBCs)

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Leukocytes

- Leukocytes (white blood cells, or WBCs) are relatively colorless compared to red blood cells.
- Are the least numerous blood cells.
- Are the cells of the immune system that are involved in protecting the body against both infectious disease and foreign invaders.
- Leukocytes are found throughout the body, including the blood and lymphatic system.
- The only blood cells that contain a nucleus. They comprise approximately 1% of the total blood volume in a healthy adult.

- The normal white cell count in adults is usually between 4,500–11,500 per cubic millimeter (mm^3) or microliter (μL) of blood (4.5×10^3 - $11.5 \times 10^3 / \mu\text{L}$), while in infants is 10,000-30,000 cells/ mm^3 (μL).
- An increase in the number of leukocytes over the upper limits is called **leukocytosis**, and a decrease below the lower limit is called **leukopenia**.
- Leukocytes develop from hematopoietic stem cells (HSCs) in the bone marrow, where most undergo differentiation and maturation, and then are released into the circulation.
- Differentiation and maturation of leukocytes occur in the bone marrow, except for T-lymphocytes which occur in the thymus.

- All WBCs are derived from the common myeloid progenitor (CFU-GEMM), except lymphocytes which are derived from the common lymphoid progenitor (CLP).
- The number of leukocytes in the blood is often an indicator of disease, and thus the **WBC count** is an important subset of the complete blood count.
- WBCs are either granulocytes or agranulocytes.
- Granulocytes are a group of leukocytes whose cytoplasm is filled with granules with differing staining characteristics and whose nuclei are segmented or lobulated.

- Granulocytes include neutrophils, eosinophils, and basophils.
- Agranulocytes include monocytes and lymphocytes.
- Monocytes and lymphocytes are also known as mononuclear cells.
- Mononuclear cells have nuclei that are not segmented but are round, oval, indented, or folded.
- Granulocytes and monocytes form phagocytes, while lymphocytes form immunocytes.
- White cells are found in the buffy coat, a thin, typically white layer of nucleated cells between the sedimented RBCS and the blood plasma.

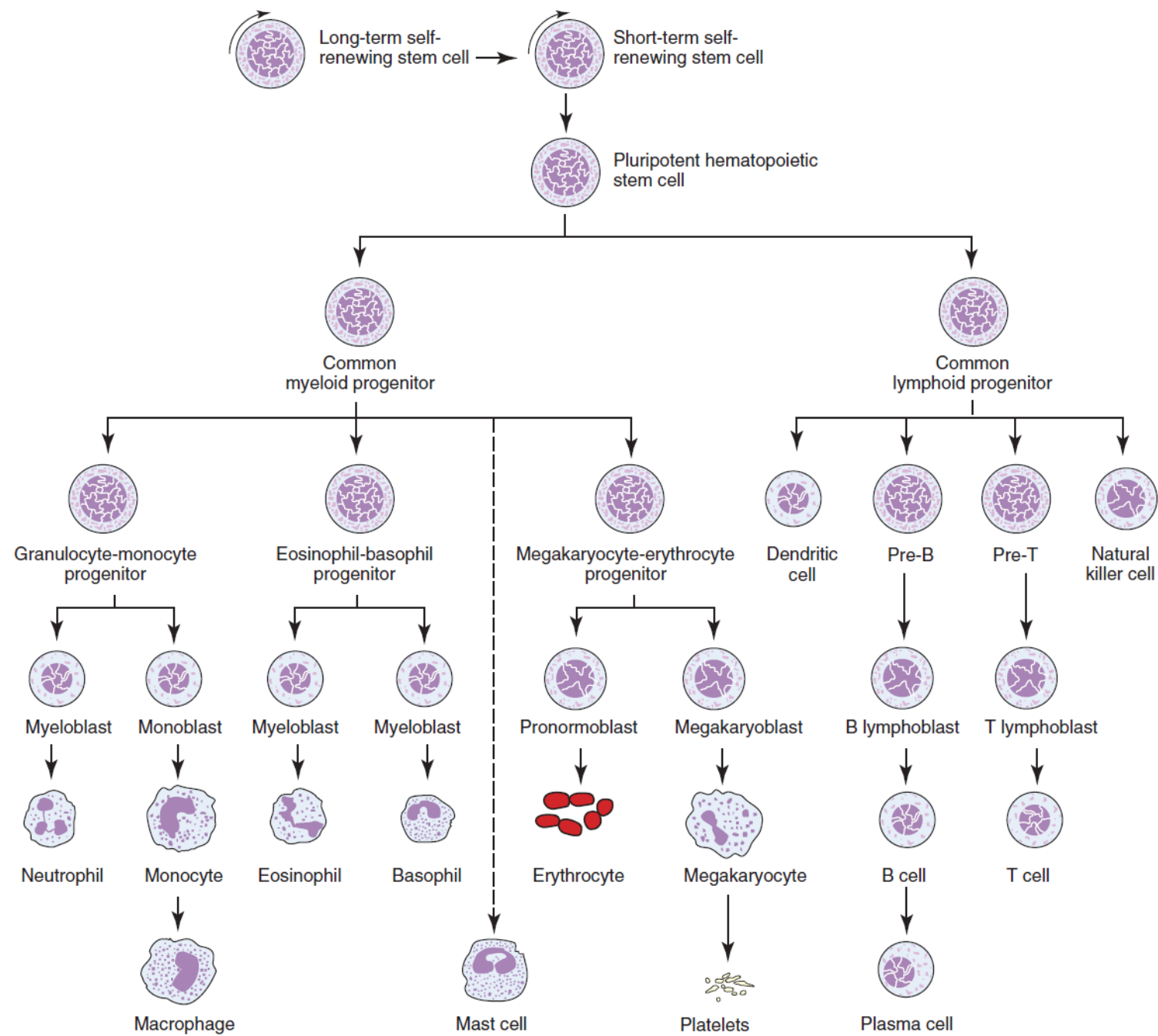
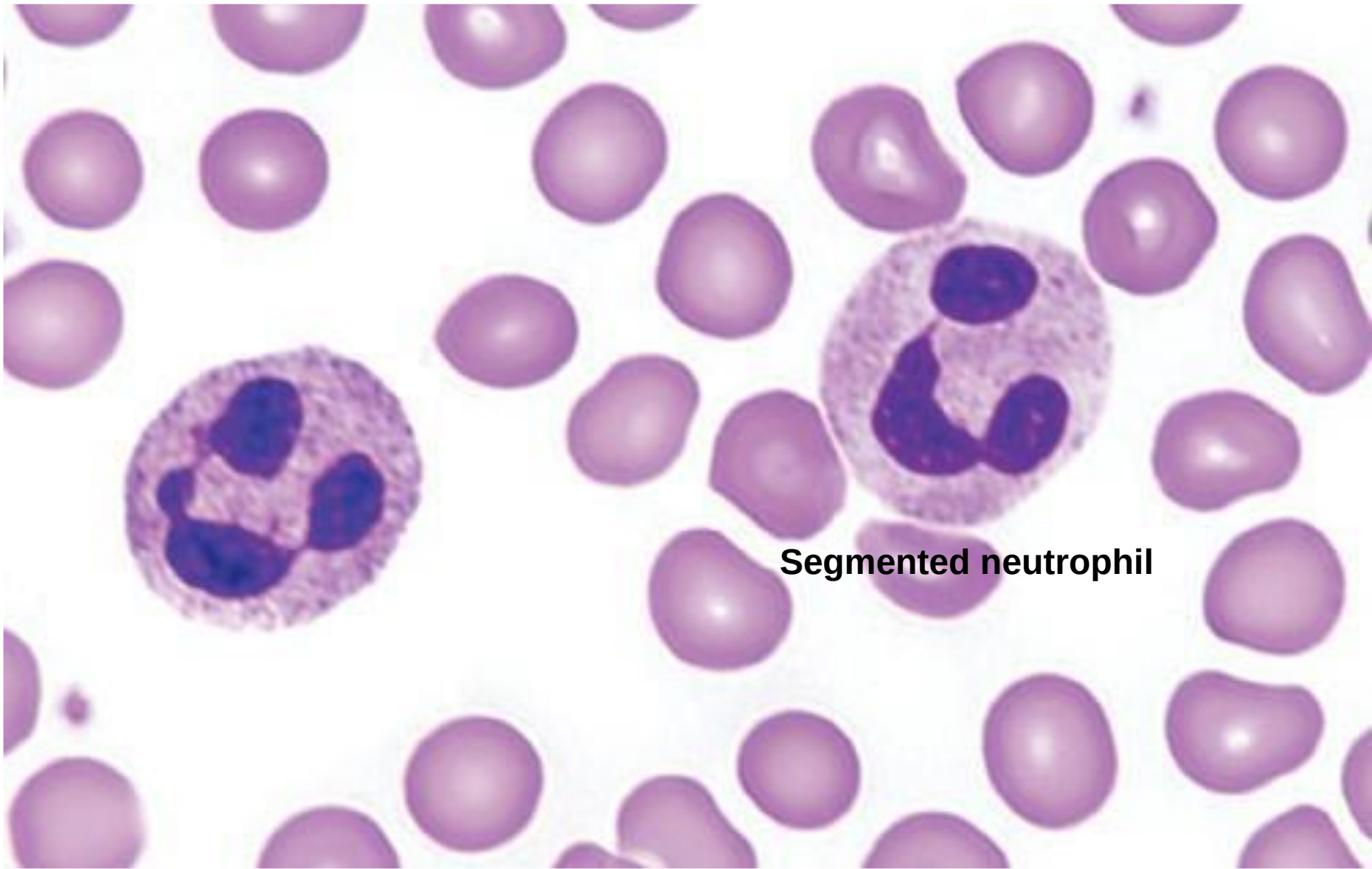


Figure 7-13 Diagram of hematopoiesis shows derivation of cells from the pluripotent hematopoietic stem cell.

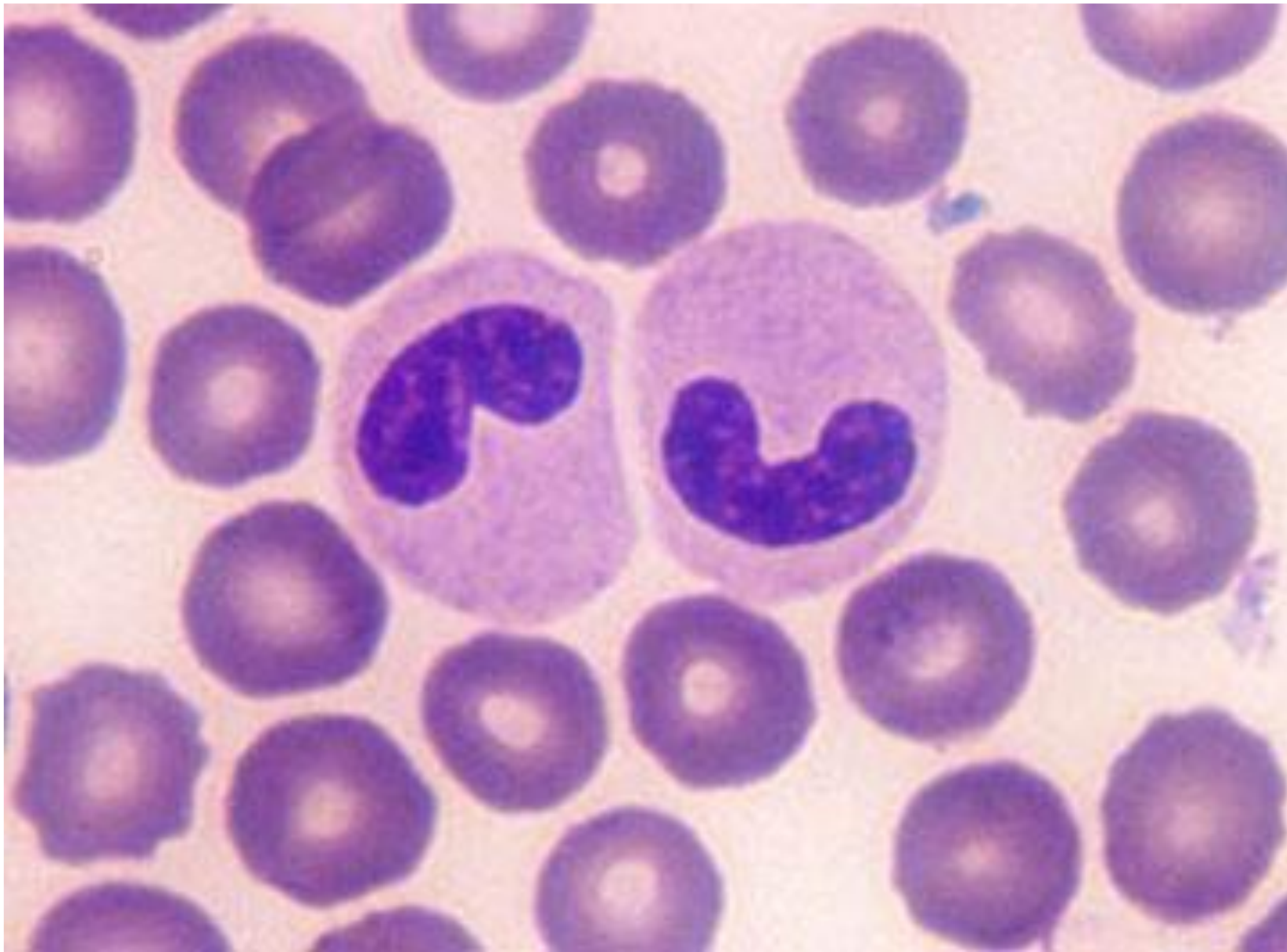
Neutrophils

- Neutrophils are the most abundant white blood cells in adults (approximately 10^{11} are produced daily).
- Constituting 50-70% of the circulating leukocytes.
- Neutrophil granulocytes have an average diameter of 12-15 μm .
- The life span of neutrophils in blood is about 7 hours.
- The name neutrophil derives from staining characteristics, whereas basophilic white blood cells stain dark blue and eosinophilic white blood cells stain bright red, neutrophils stain a neutral pink.
- They have a multi-lobed nucleus, which consists of three to five lobes connected by slender strands.

- Neutrophils may be subdivided into segmented neutrophils and band neutrophils (or bands, stab or juvenile).
- Hypersegmentation (more than 5 lobes) is not normal and occurs in some disorders, most notably Vitamin B12 deficiency.
- The major function is phagocytosis and destruction of foreign material and microorganisms (bacteria).
- They are usually the first responders to microbial infection; their activity and death in large numbers form pus.
- Neutrophils are active in phagocytizing bacteria and are present in large amounts in the pus of wounds (die after having phagocytosed a few pathogens).



Segmented neutrophil

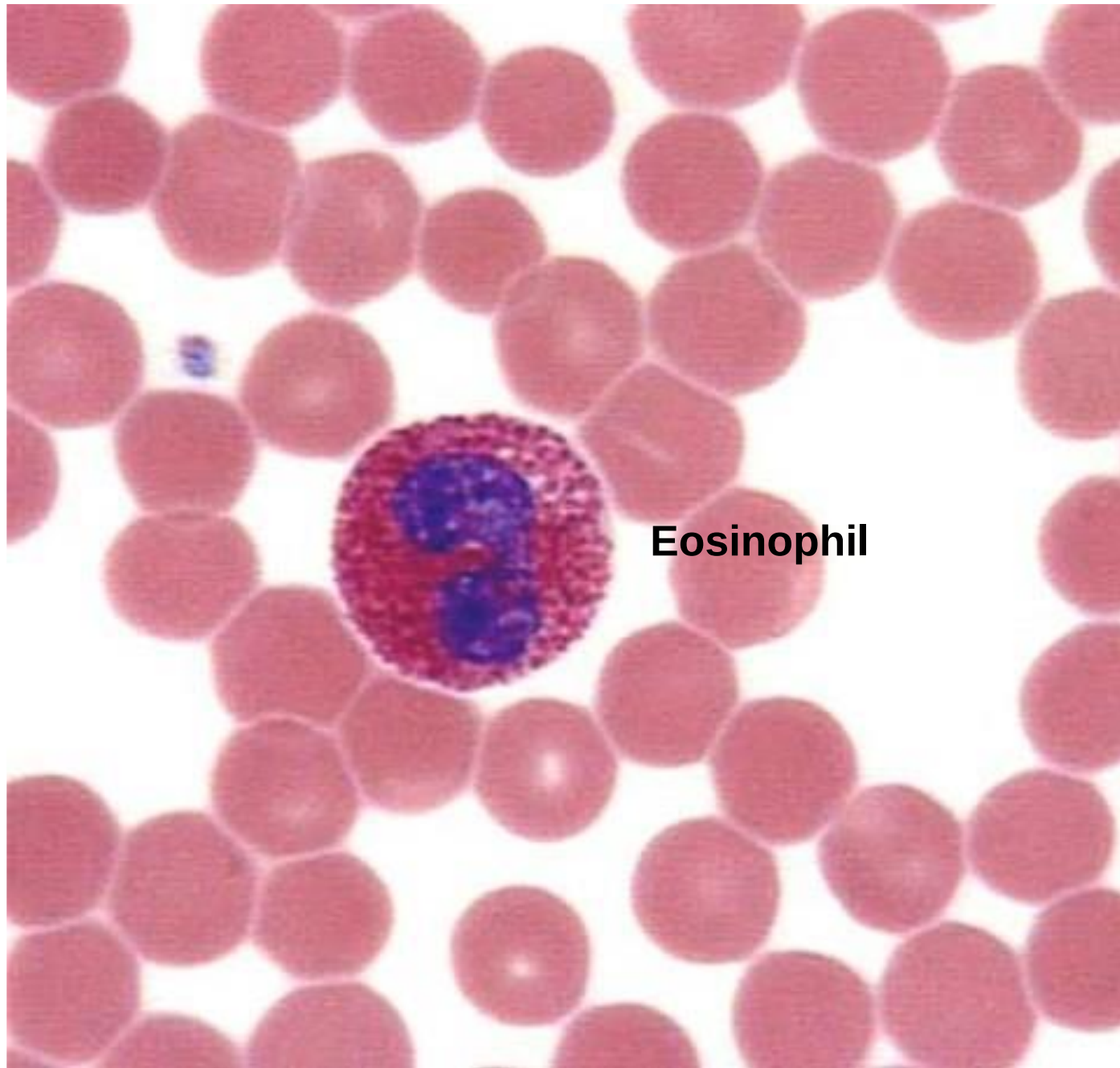


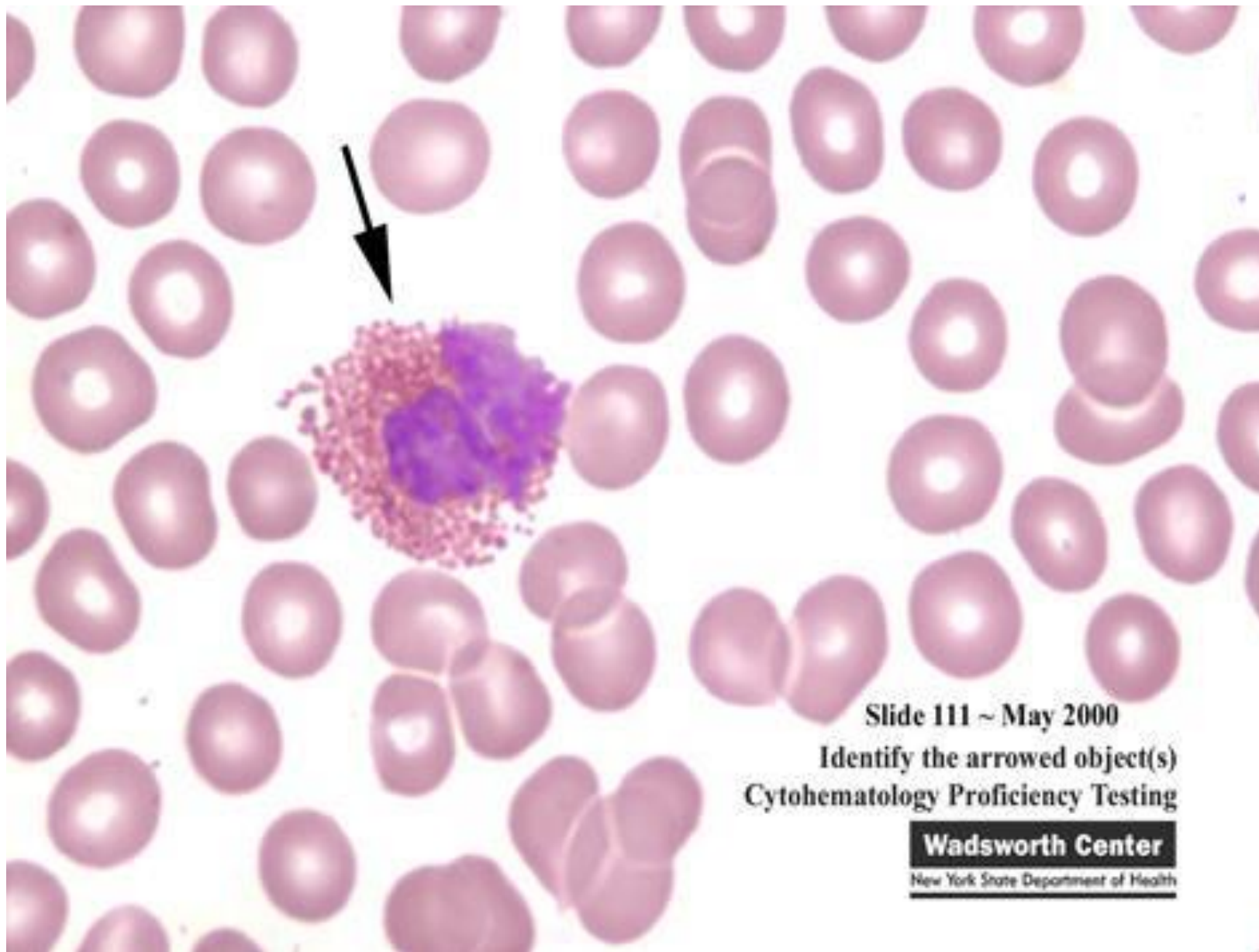
Band neutrophil

Eosinophils

- Eosinophils compose about 1-3% of the WBC total.
- Eosinophils have an average diameter of 12-17 μm .
- Eosinophils persist in the circulation for 18 hours
- Their nucleus is bilobed. The lobes are connected by a thin strand.
- The cytoplasm is full of granules that assume a characteristic orange-red color with eosin staining.

- They primarily fight against parasitic infections (helminths).
- Eosinophilia is a hallmark of allergic reactions (asthma).
- They secrete chemicals that destroy these large parasites (**degranulation**), that are too big for any one WBC to phagocytize.





Slide 111 ~ May 2000

Identify the arrowed object(s)
Cytohematology Proficiency Testing

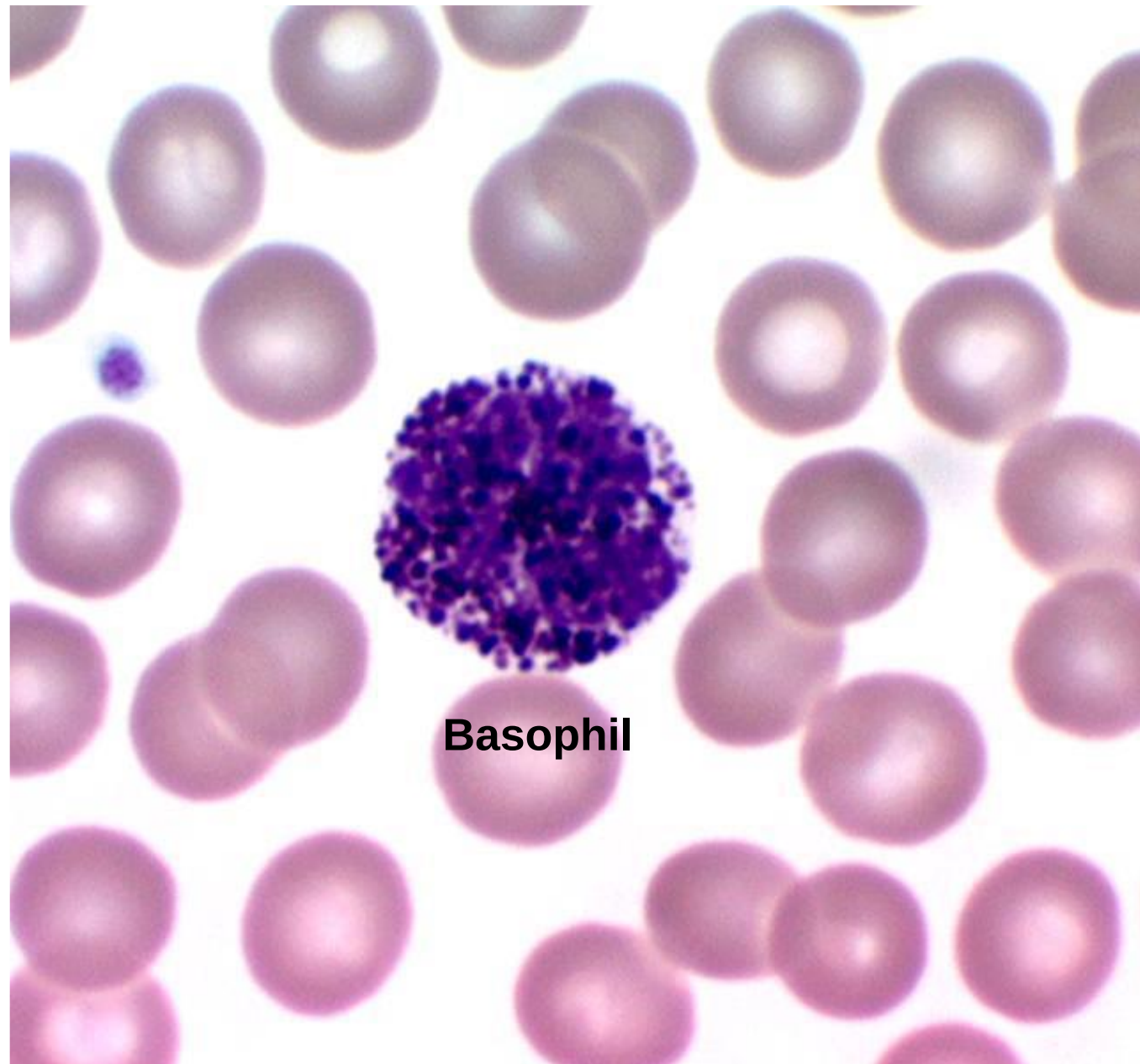
Wadsworth Center

New York State Department of Health

Eosinophil degranulation

Basophils

- Are the least numerous WBCs (0-2%).
- The nucleus is bi-lobed and difficult to be seen.
- Contain dark cytoplasmic granules, which obscure the nucleus and contain heparin and histamine.
- Have a life span of 60 hours.
- Like eosinophils, basophils play a role in both parasitic infections and allergies.
- They have IgE receptors and their degranulation is associated with histamine release.

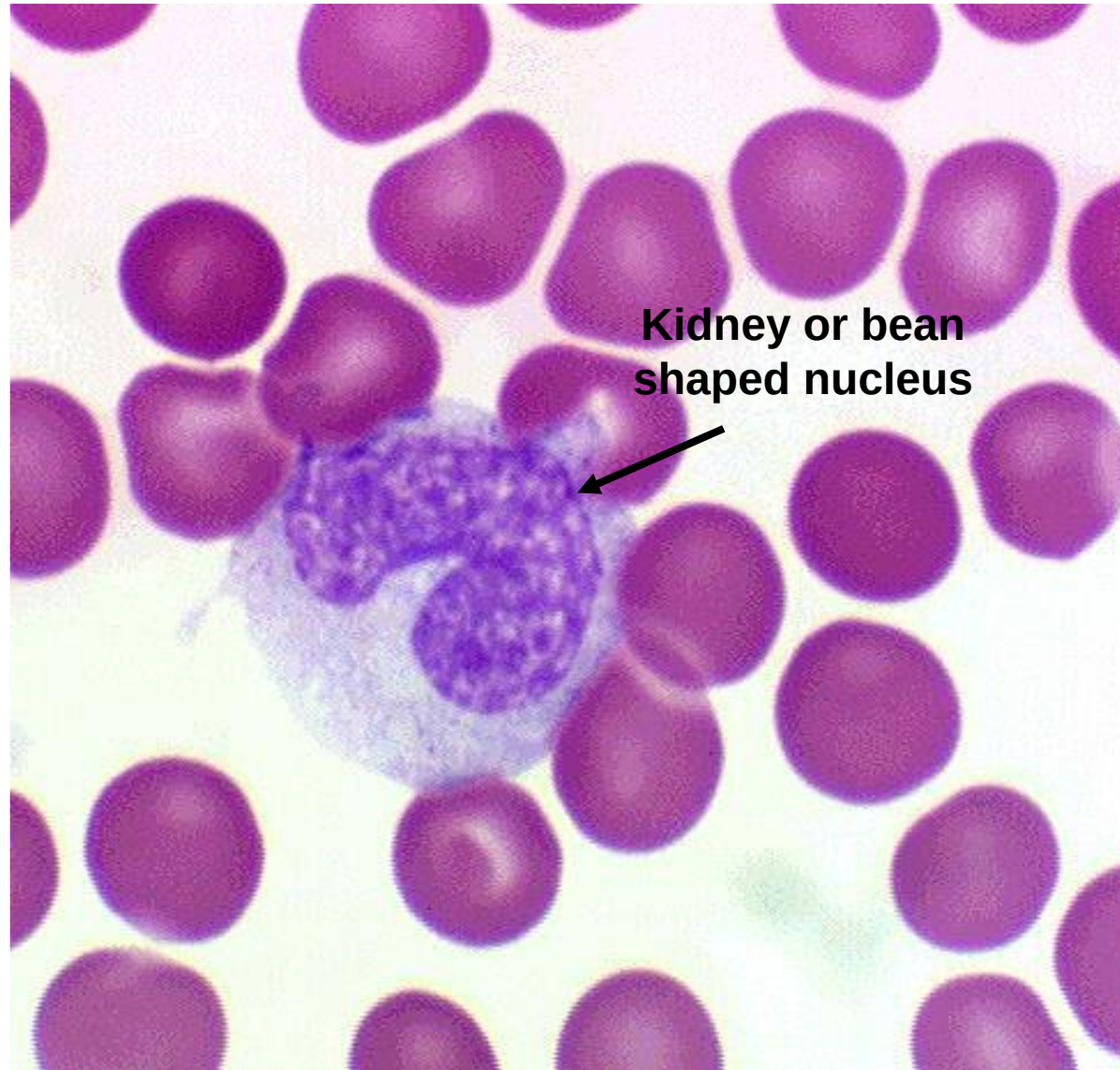


Basophil

Monocytes

- The largest type of WBCs (15-20 μm).
- Monocytes constitute 2% to 11% of circulating leukocytes.
- Monocytes have unilobar nuclei, which makes them one of the types of mononuclear leukocytes.
- Monocytes are slightly immature cells whose ultimate goal is to enter the tissues and mature into macrophages, osteoclasts, or dendritic cells.
- The nucleus may be round, oval, or kidney-shaped, but more frequently is deeply indented (horseshoe-shaped) or folded on itself.

- Half of them are stored in the spleen.
- Monocytes remain in circulation for approximately 3 days.
- They have many functions: phagocytosis of foreign substances, destruction of aged RBCs and maintenance of storage pool of iron, antigen-presenting to the T cell, and removal of dead cells and debris.
- Share the “vacuum cleaner” (phagocytosis) function of neutrophils, but are much longer lived as they have an extra role: they present pieces of pathogens to T cells so that the pathogens may be recognized again and killed.



Lymphocytes

- The 2nd most abundant WBCs in adults (18-42%).
- Lymphocytes are much more common in the lymphatic system than in blood.
- Lymphocytes are distinguished by having a deeply staining nucleus that may be eccentric in location, and a relatively small amount of cytoplasm.
- The nucleus of a lymphocyte is approximately the size of a red blood cell (about 7.0 μm in diameter).
- They are the cells of the adaptive immunity.

- The function of T cells and B cells is to recognize specific “non-self” antigens, during a process known as antigen presentation.
- It is impossible to distinguish between T cells and B cells in a peripheral blood smear.
- Lymphocytes may be small or large.

Lymphocytes include:

1. B cells:

- Play a major role in the antibody-mediated or humoral immunity.
- The differentiation and maturation occur in the BM.
- They differentiate to give **plasma cells (effector cells)** that make antibodies that can bind to pathogens, block pathogen invasion, activate the complement system, and enhance pathogen destruction **and memory cells.**
- Represent 3-21% of circulating lymphocytes.

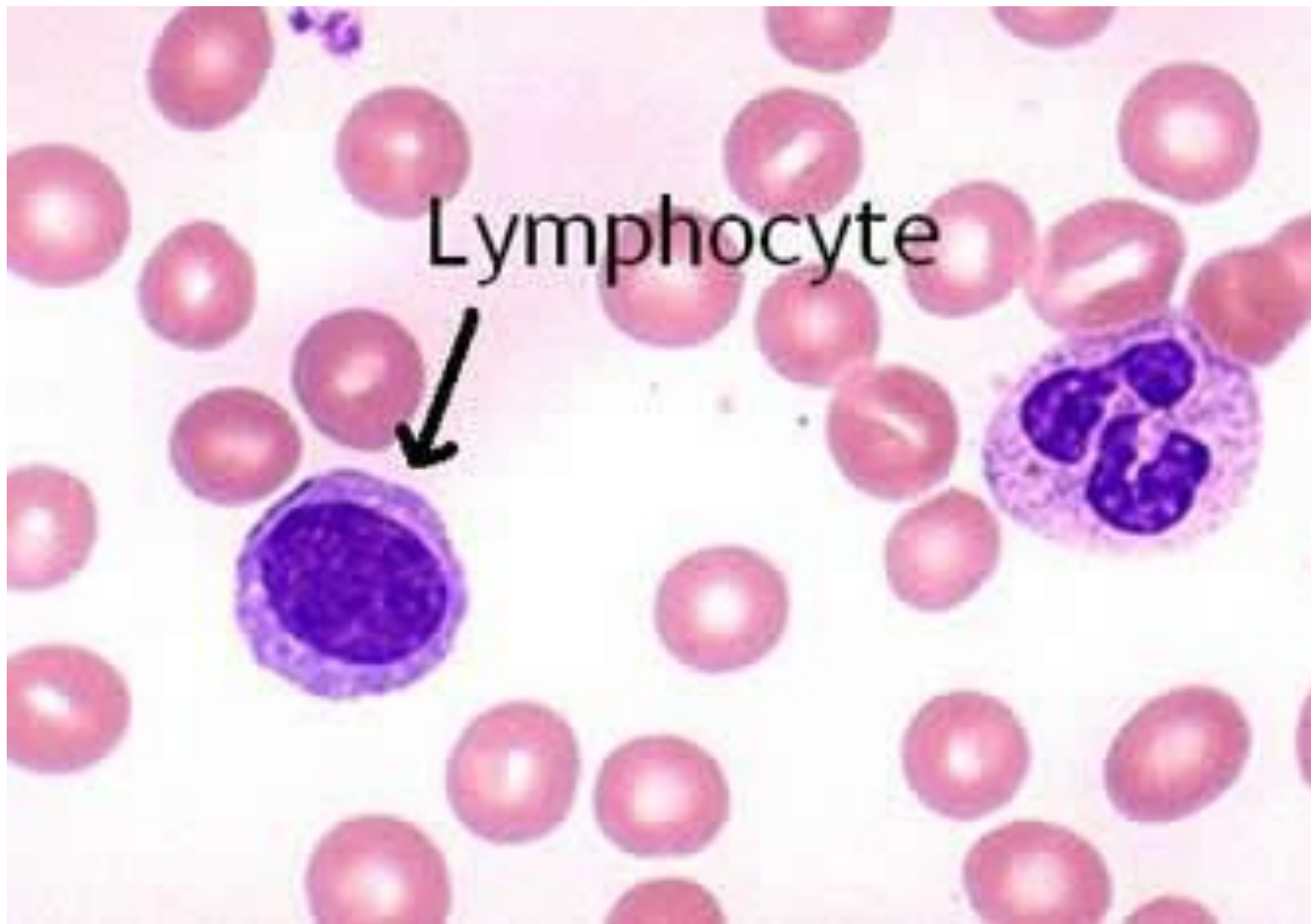
2. **Natural killer (NK) cells:** function as part of innate immunity and are capable of killing certain tumor cells and virus-infected cells. Approximately 4-29% of circulating lymphocytes are NK cells.

3. T cells:

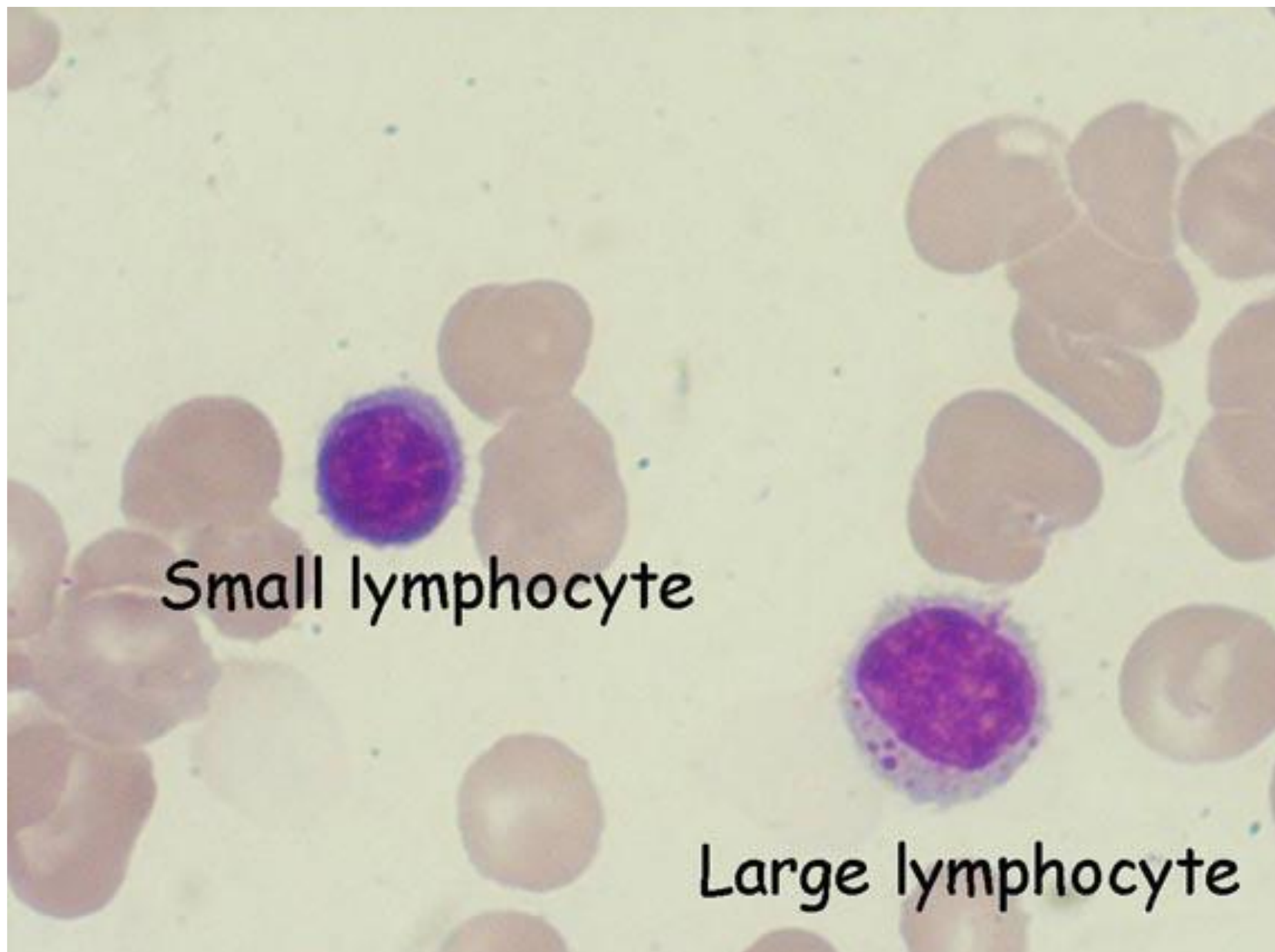
- They play a major role in the cell-mediated or cellular immunity.
- T lymphocytes develop in the thymus.
- Represent about 51-88% of circulating lymphocytes.

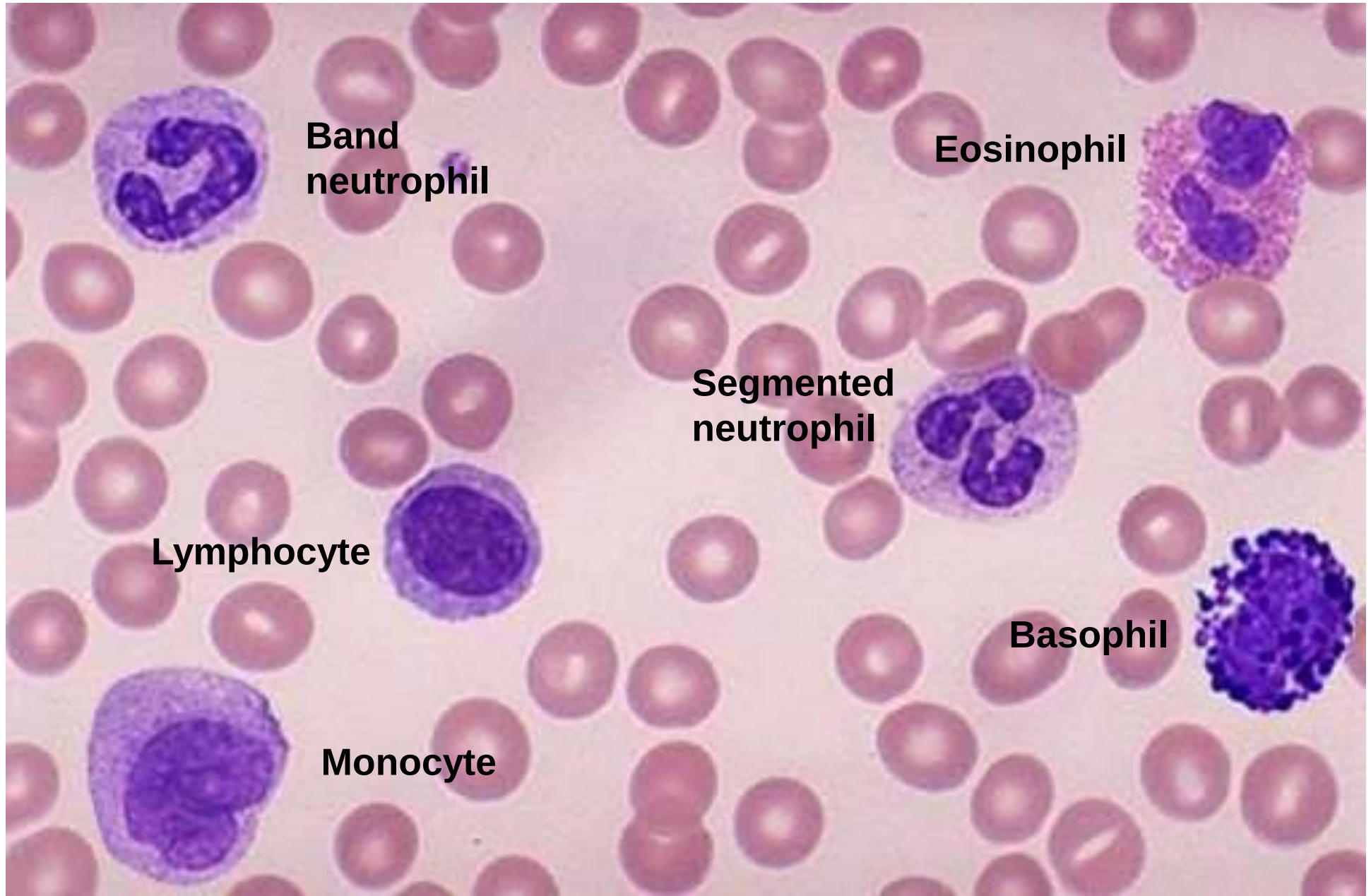
T-cells are sub-divided into:

- 1. CD4+ helper T cells:** known as CD4+ T cells. These cells have T-cell receptors and CD4 molecules. Helper T cells make cytokines and perform other functions that help coordinate the immune response. **These cells are the regulator cells of the immune system.**
 - In HIV infection, these T cells are attacked by the virus.
- 2. CD8+ cytotoxic T cells:** known as CD8+ T cells, are capable of killing target cells.



**This lymphocyte has deeply staining nucleus
and small cytoplasm (rim)**





White blood cells (Leukocytes)

